

Lecture 12: Higher Order Derivatives

GOAL: Compute derivatives of derivatives. Interpret the second and third derivative in a physics context.

Summary:

$$\frac{d^n f}{dx^n} = \frac{d}{dx} \left(\frac{d^{n-1} f}{dx^{n-1}} \right) \quad f^{(n)}(x) = [f^{(n-1)}(x)]'$$
$$a(t) = v'(t) = s''(t) \quad j(t) = a'(t) = v''(t) = s'''(t)$$

Recall that the height of a ball tossed straight into the air can be modeled by a parabola:

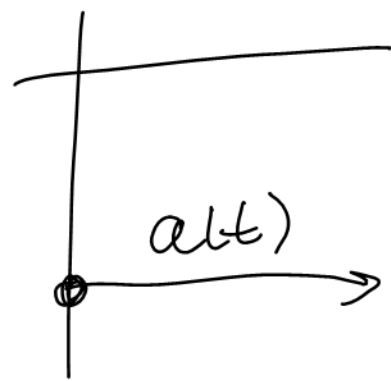
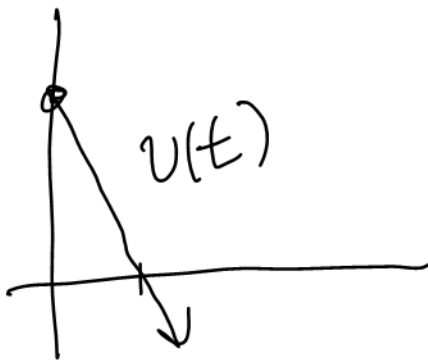
$$s(t) = -4.9t^2 + 6t + 7$$

The velocity:

$$v(t) = s'(t) = -9.8t + 6$$

$v(t)$ is differentiable, so can keep going

$$v'(t) = -9.8 \stackrel{\text{def}}{=} a(t)$$



def The quantity $\frac{d}{dx} \left(\frac{df}{dx} \right) = \frac{d^2 f}{dx^2}$ is called the 2nd Derivative of f w.r.t. x

$$[f'(x)]' = f''(x)$$

Example 1 Find $f''(x)$ if $f(x) = 2 \sin x + 3 \cos x$

$$f(x) = 2 \sin x + 3 \cos x$$

$$f'(x) = 2 \cos x - 3 \sin x$$

$$\begin{aligned} f''(x) &= [f'(x)]' = -2 \sin x - 3 \cos x \\ &= -(2 \sin x + 3 \cos x) = -f(x) \end{aligned}$$

Def The acceleration of an object, whose position at time t is given $s(t)$, is defined by $a(t) \stackrel{\text{def}}{=} v'(t) = [s'(t)]' = s''(t)$

Example 2 A particle is traveling along a straight line; its position at time t is given by

$$s(t) = t^3 - 6t^2 + 9t$$

a) Find the velocity and acceleration at time t .

$$v(t) = s'(t) = 3t^2 - 12t + 9$$

$$a(t) = [v'(t)]' = 6t - 12$$

b) If t is measured in seconds and $s(t)$ is measured in meters, find $v(4)$ and $a(4)$.

$$v(4) = 3(4)^2 - 12(4) + 9 = 9 \frac{m}{s}$$

$$a(4) = 6(4) - 12 = 12 \frac{\text{m/s}}{\text{s}} \\ = 12 \frac{\text{m}}{\text{s}^2}$$

c) What is the acceleration when the velocity is $24 \frac{\text{m}}{\text{s}}$?

Step 1 Find the time when $v(t) = 24$
 $v(t) = 3t^2 - 12t + 9$ set 24

$$3t^2 - 12t - 15 = 0$$

$$t^2 - 4t - 5 = 0$$

$$(t+1)(t-5) =$$

$$\Rightarrow t = -1, 5$$

As $t \geq 0$, $v(t) = 24$ when $t = 5$.

Step 2 Find $a(t)$ when $t = 5$

$$a(t) = 6t - 12$$

$$a(5) = 6(5) - 12 = 18 \frac{\text{m}}{\text{s}^2}$$

Example 3 Suppose the position of a particle is given by

$$s(t) = \frac{t}{t^2 + 1}$$

$$(t \geq 0)$$

When is the acceleration 0?

$$v(t) = \frac{(1)(t^2 + 1) - t(2t)}{(t^2 + 1)^2} = \frac{t^2 + 1 - 2t^2}{(t^2 + 1)^2}$$

$$v(t) = \frac{1-t^2}{(t^2+1)^2} = -\frac{t^2-1}{(t^2+1)^2}$$

$$a(t) = \frac{(-2t)(t^2+1)^2 - (1-t^2)(2)(t^2+1)(2t)}{(t^2+1)^4}$$

$$= \frac{-2t(t^2+1)[(t^2+1) + 2(1-t^2)]}{(t^2+1)^4}$$

$$= \frac{-2t(t^2+1)[3-t^2]}{(t^2+1)^4} \quad \underline{\text{set } 0}$$

Never 0

$$-2t = 0 \quad \text{OR} \quad t^2+1 = 0 \quad \text{OR} \quad 3-t^2 = 0$$

Never

$t^2 = 3$
 $t = \pm\sqrt{3}$

$$\xrightarrow{t \geq 0} \boxed{t = \sqrt{3}}$$

Higher-Order Derivatives

If the derivative is diff. then there's nothing stopping from taking the derivative

$$\text{3rd:} \quad \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right) = \frac{d^3 f}{dx^3}$$

$$[f''(x)]' = f'''(x)$$

$$\text{4th:} \quad \frac{d}{dx} \left(\frac{d^3 f}{dx^3} \right) = \frac{d^4 f}{dx^4}$$

n^{th}

$$\begin{aligned} [f'''(x)]' &= f''''(x) \\ \frac{d}{dx} \left(\frac{d^{n-1}f}{dx^{n-1}} \right) &\stackrel{\text{def}}{=} \frac{d^n f}{dx^n} \\ [f^{(n-1)}(x)]' &\stackrel{\text{def}}{=} f^{(n)}(x) \end{aligned} \quad \left| \begin{aligned} f^{(0)}(x) &= f(x) \\ f^{(1)}(x) &= f'(x) \end{aligned} \right.$$

NOTE: The third derivative of a position fun $s(t)$ is called the jerk at t , denoted $j(t) \stackrel{\text{def}}{=} a'(t) = v''(t) = s'''(t)$

Example 4
If $f^{(9)}(x) = 6 \sec(2x-5)$, find $f^{(10)}(x)$.

$$\begin{aligned} f^{(10)}(x) &= [f^{(9)}(x)]' = [6 \sec(2x-5)]' \\ &= 6 \sec(2x-5) \tan(2x-5) (2) \\ &= 12 \sec(2x-5) \tan(2x-5) \end{aligned}$$

Example 5 Examine the various derivatives of $\sin x$

$$\begin{aligned} f(x) &= \sin x \\ f'(x) &= \cos x \\ f''(x) &= -\sin x \\ f'''(x) &= -\cos x \end{aligned} \quad \rightarrow f^{(4)}(x) = \sin x$$

```

    graph TD
      A[sin x] --> B[cos x]
      B --> C[-sin x]
      C --> D[-cos x]
      D --> A
  
```

Example 6 Repeat for $f(x) = x^3 + x^2 + x + 1$

$$\boxed{x^3 + x^2 + x + 1}$$

$\downarrow \frac{d}{dx}$

$$\boxed{3x^2 + 2x + 1}$$

$\downarrow$

$$\boxed{f'(x) = 6x + 2}$$

$\downarrow$

$$\boxed{f''(x) = 6}$$

$\downarrow$

$$\boxed{0}$$

Example 7

Repeat for $f(x) = e^{2x}$

$$f(x) = e^{2x}$$

$$f'(x) = 2e^{2x}$$

$$f''(x) = 2e^{2x}(2) = 2^2 e^{2x}$$

$$f'''(x) = 2^2 e^{2x}(2) = 2^3 e^{2x}$$

$$f^{(n)}(x) = \dots \longrightarrow = 2^n e^{2x}$$

Ex 8 (HW 12, 2) Find $g''(x)$ if

$$g(x) = 4e^{3x} \cos(4x)$$

$$g'(x) = 4e^{3x} (3) \cos(4x) + 4e^{3x} (-\sin(4x)) (4) \\ = 4e^{3x} [3 \cos(4x) - 4 \sin(4x)]$$

$$g''(x) = 12e^{3x} [3 \cos(4x) - 4 \sin(4x)] \\ + 4e^{3x} [-12 \sin(4x) - 4 \cos(4x) (4)] \\ = -28e^{3x} \cos(4x) - 96e^{3x} \sin(4x)$$